MOCCA 2025
Fundamentals of Character Animation and Motion Control

Lecture 12:

Muscle-actuated Characters

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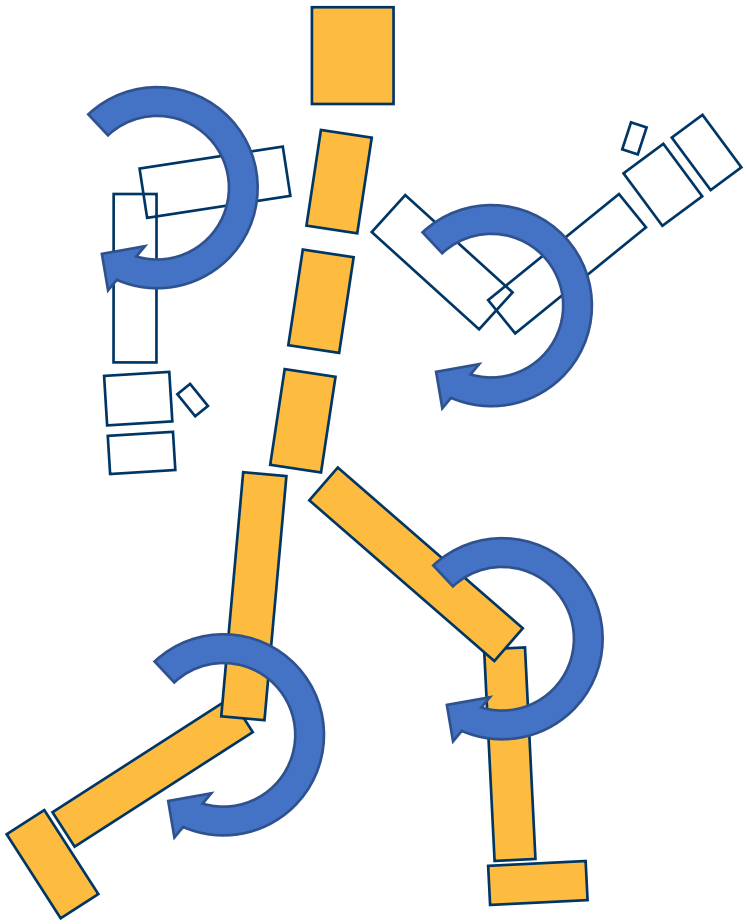
Outline

- Musculoskeletal Systems
  - Hill-type muscle model
  - Control with musculoskeletal systems

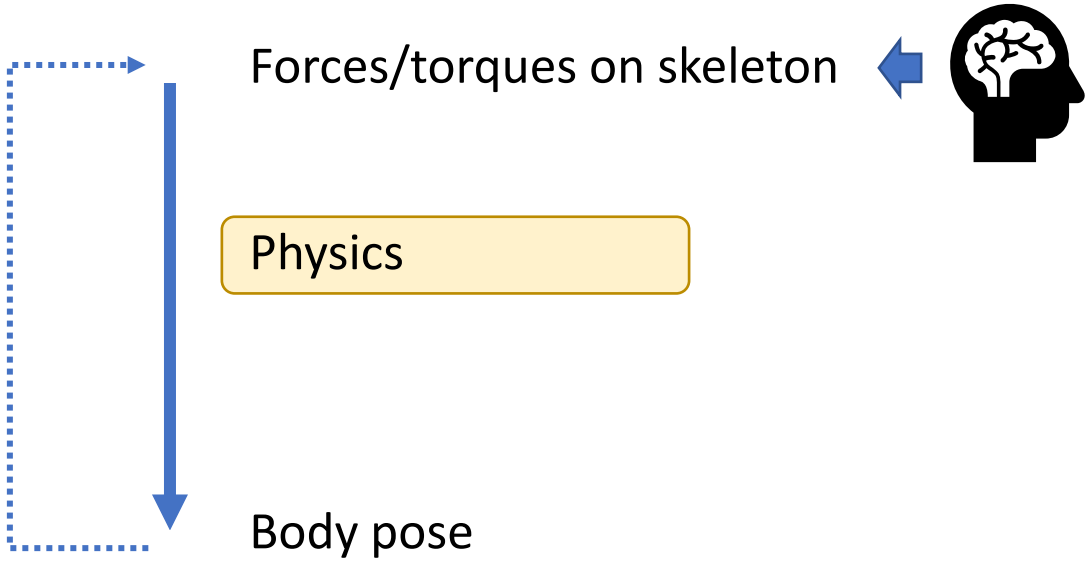
Where does a motion come from?



Neuronal excitation

Muscle activation

Forces/torques on skeleton



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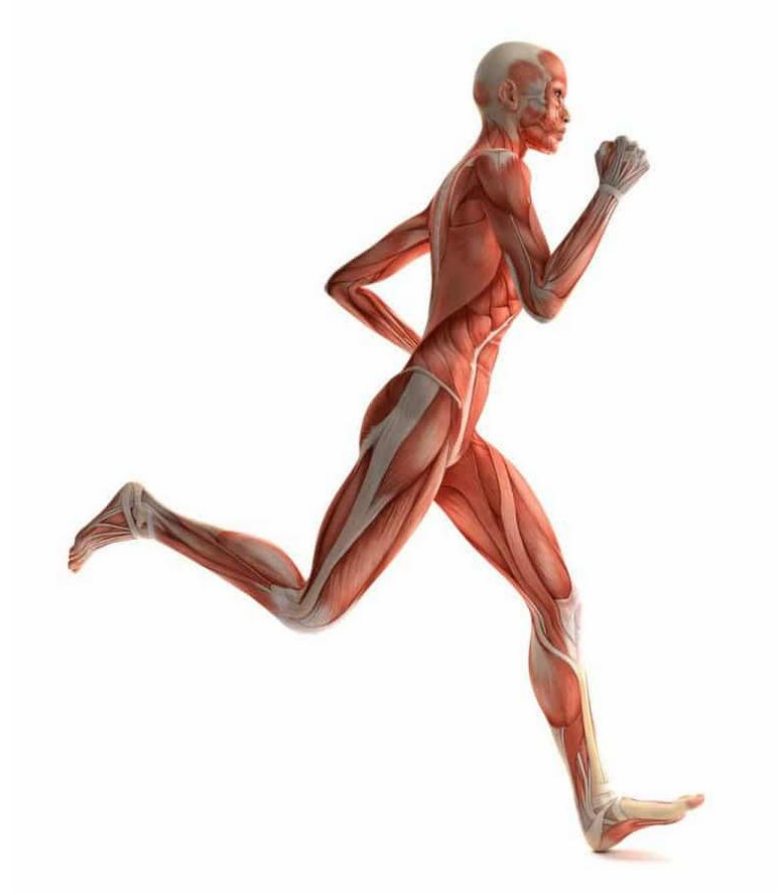

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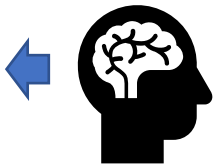


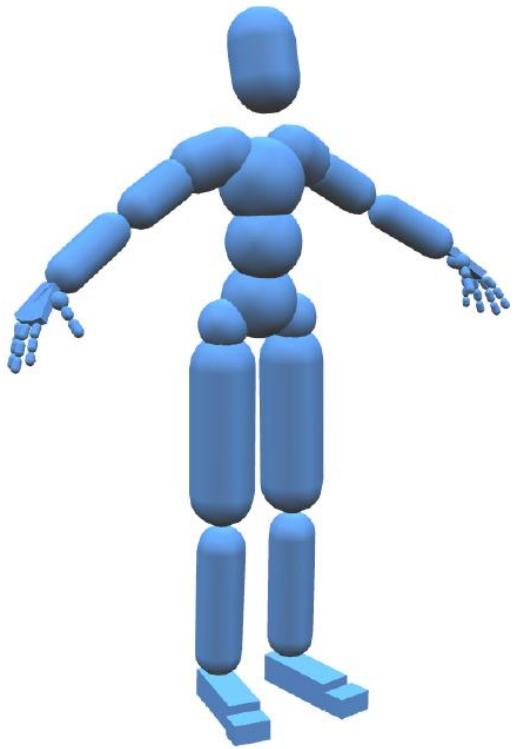
Muscle activation

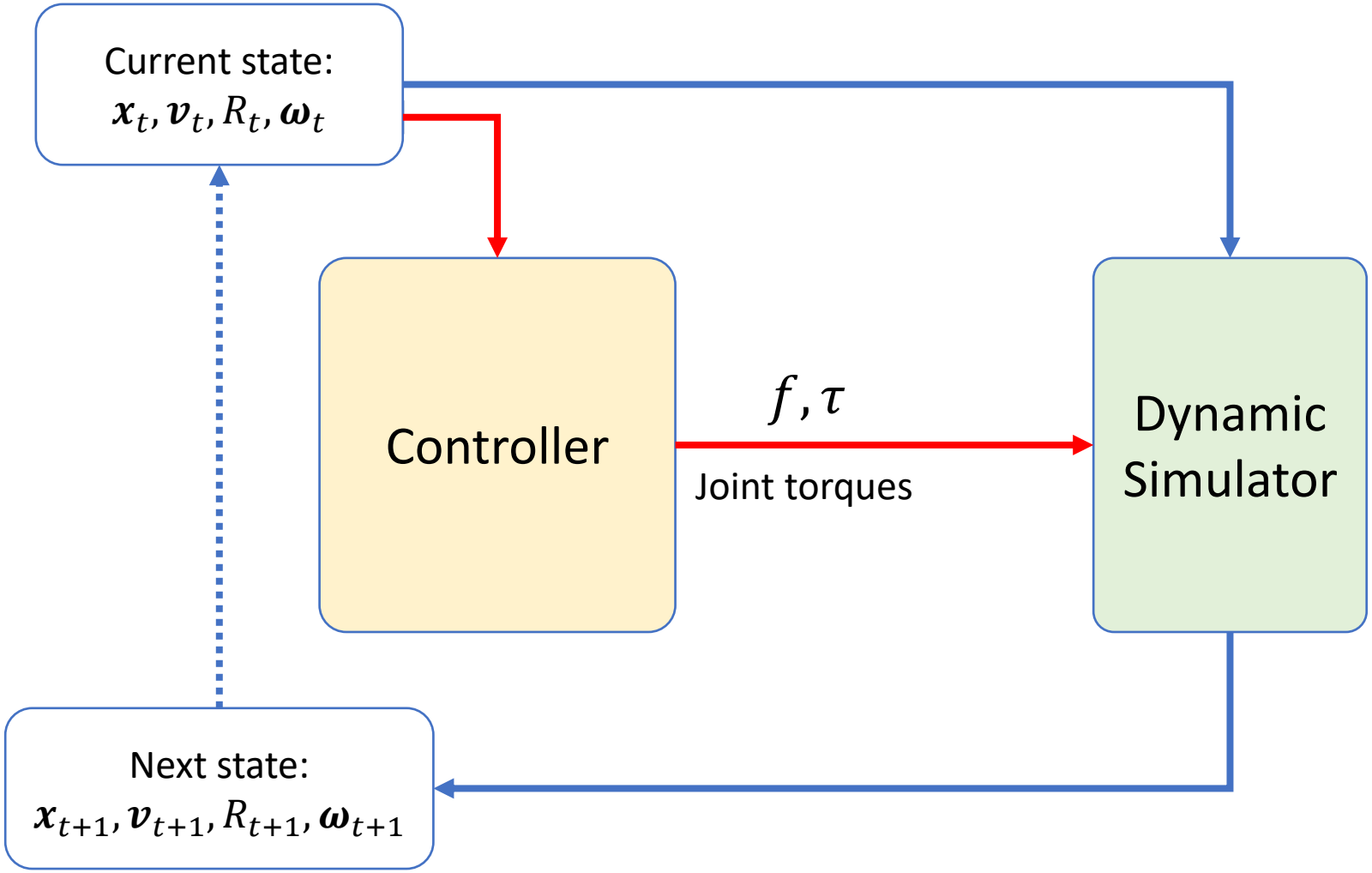
Forces/torques on musculoskeletal system

Physics

Body pose

Simulating + Controlling a Character





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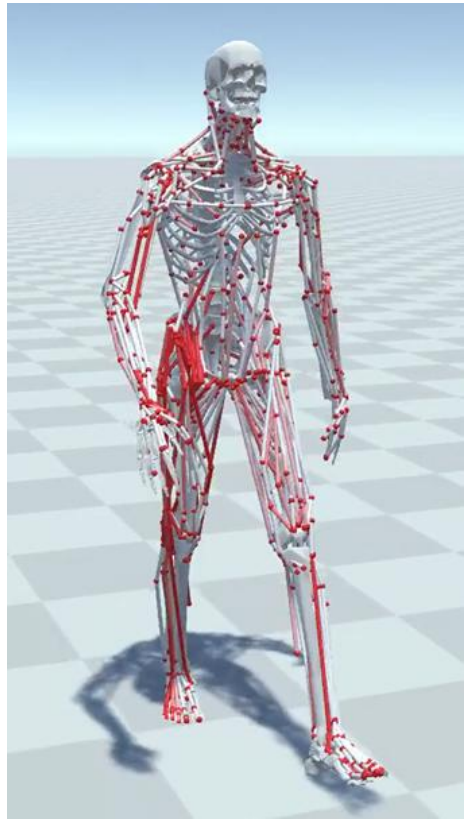
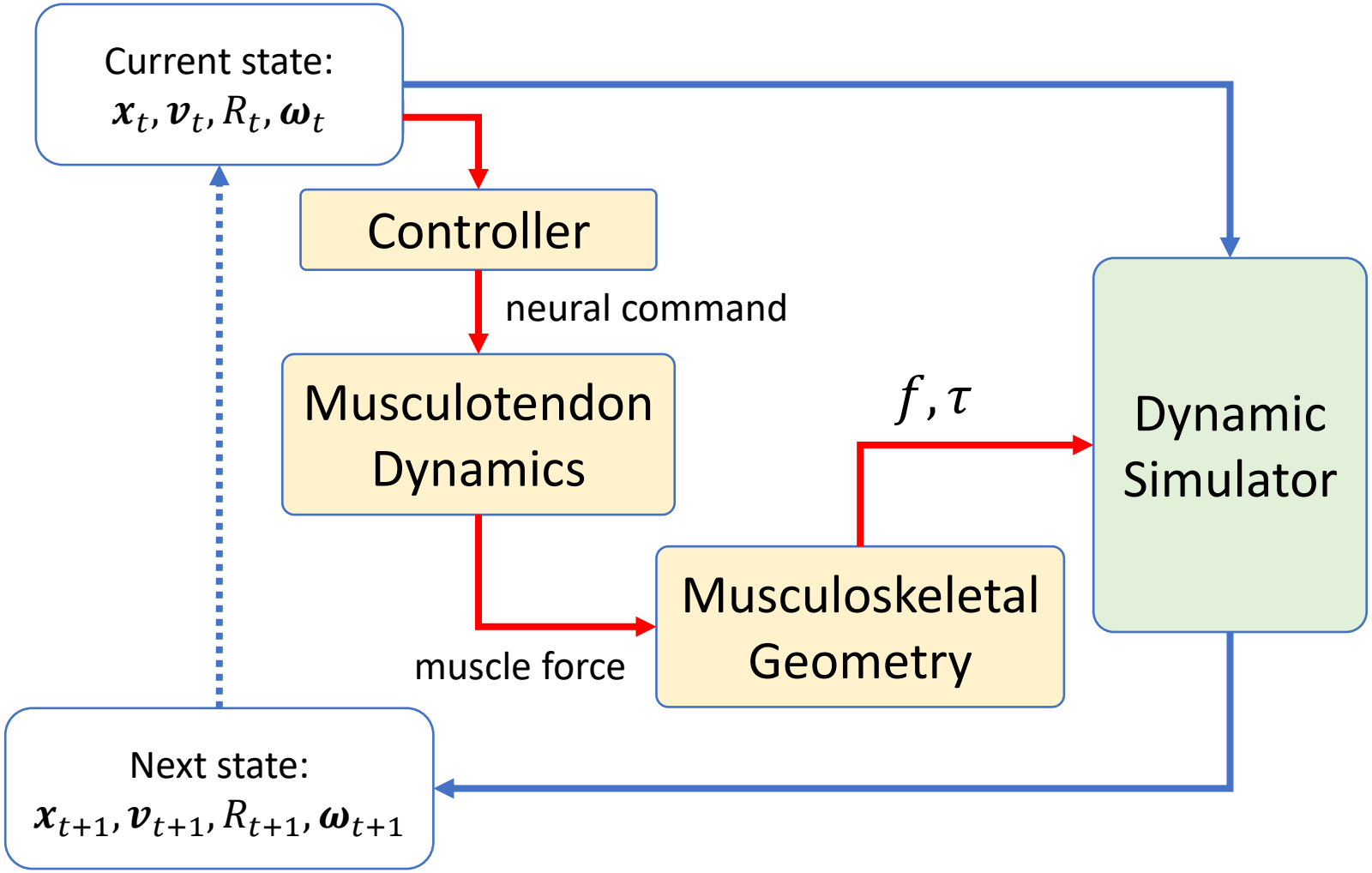
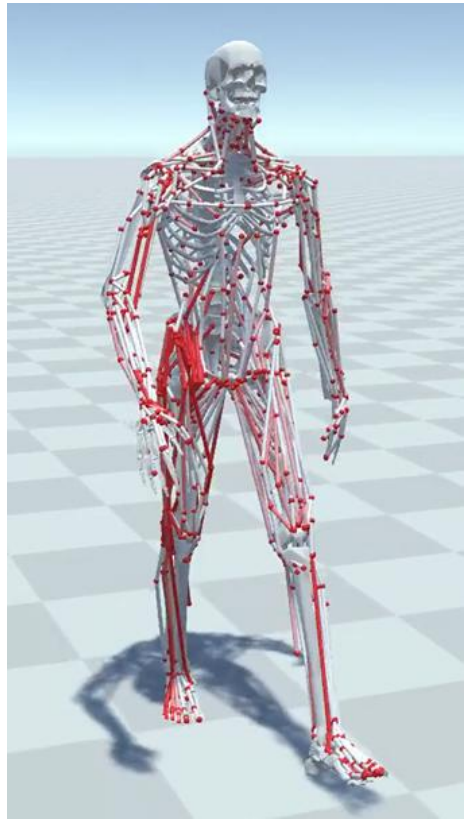
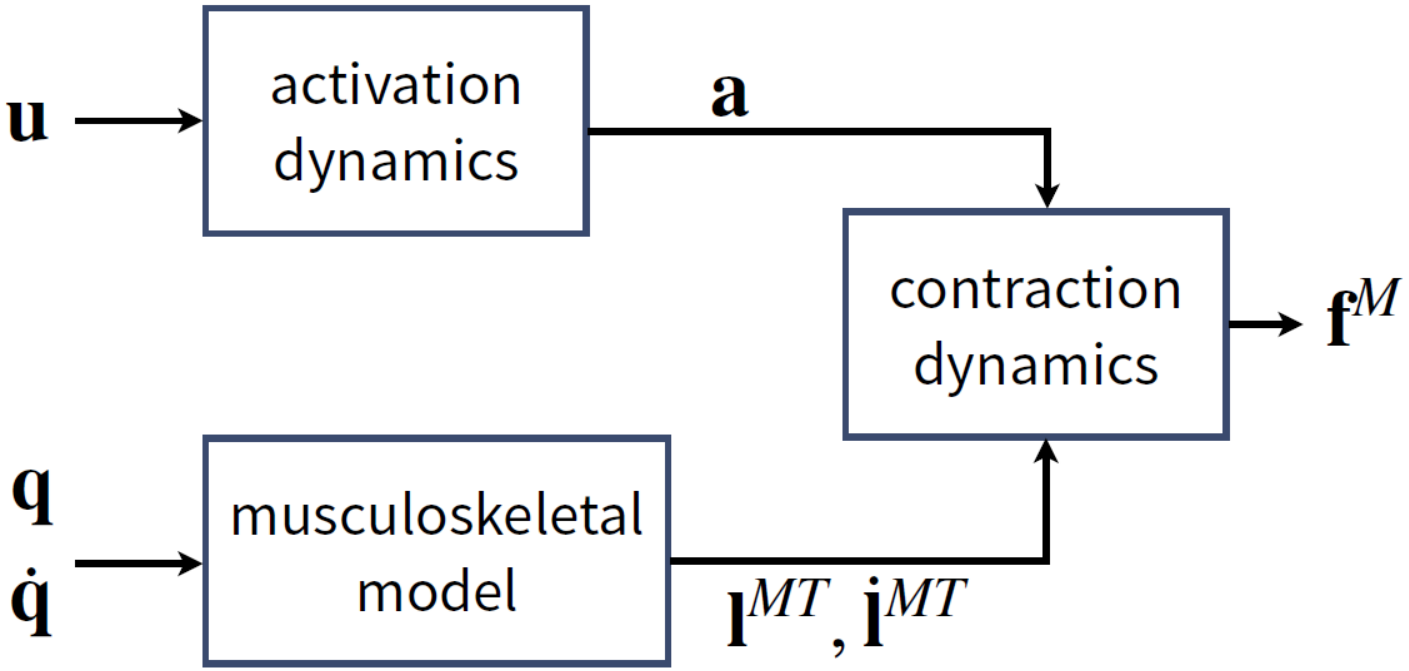




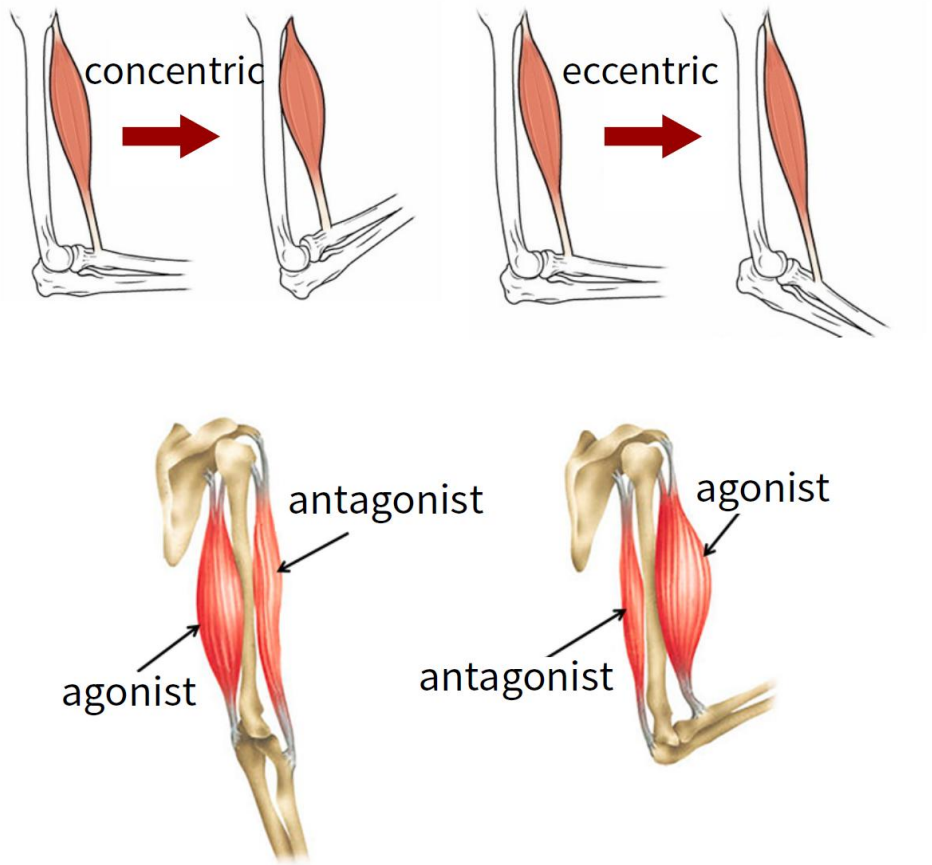
Musculotendon Dynamics





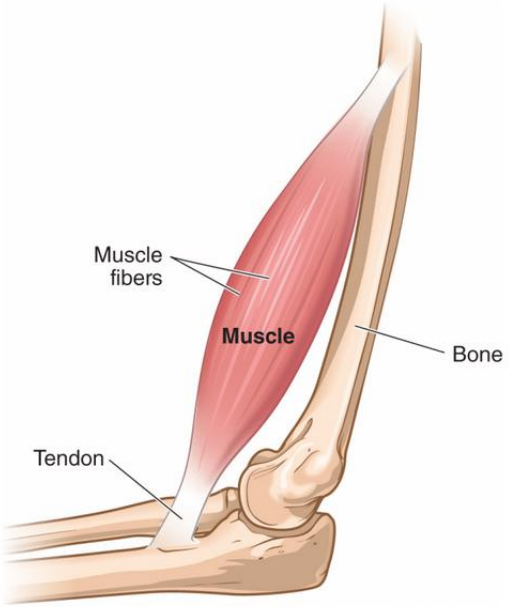
Muscle Contraction

- Muscles can only produce force through contraction
  - includes both shortening (concentric contraction) and lengthening (eccentric contraction).
- Muscles operate in 'antagonistic pairs,' meaning when one muscle contracts, the other one relaxes or lengthens.
  - The 'agonist' is the muscle undergoing concentric contraction.
  - The 'antagonist' is the muscle that is relaxing or experiencing eccentric contraction.

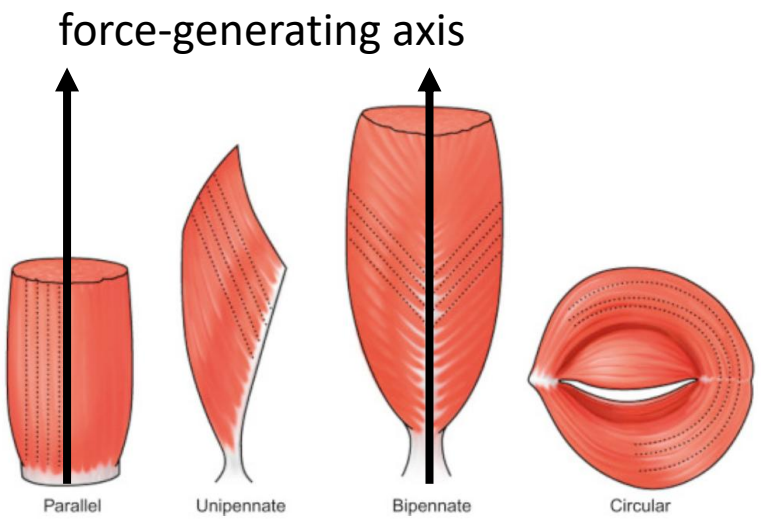


Muscle and Tendon

- Each 'muscle' is modeled as a Muscle-Tendon Unit (MTU), which consists of an active muscle fiber and a passive tendon in series.
- The abstract 3D muscle volume is represented by a massless, 1D centroidal path, which signifies both the muscle and the tendon.
- The internal architecture of a muscle profoundly affects the amount and direction of tension developed by contracting muscle fibers.



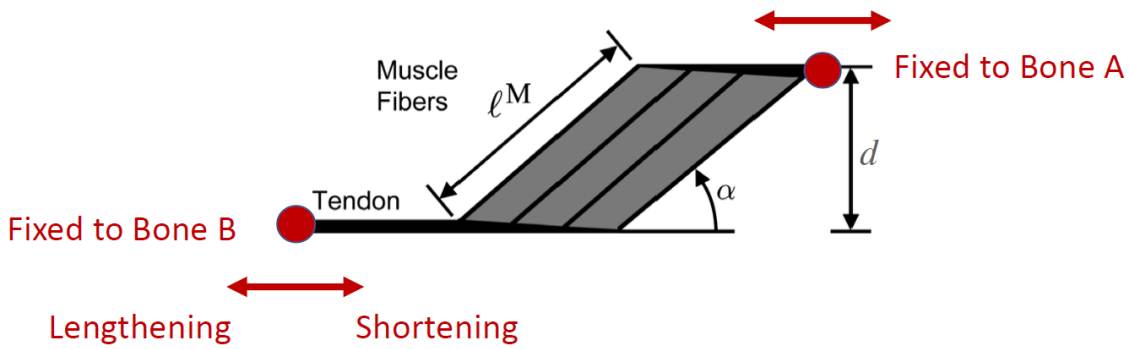
Bone, Muscle and Tendon



Muscle-Tendon Unit (MTU)

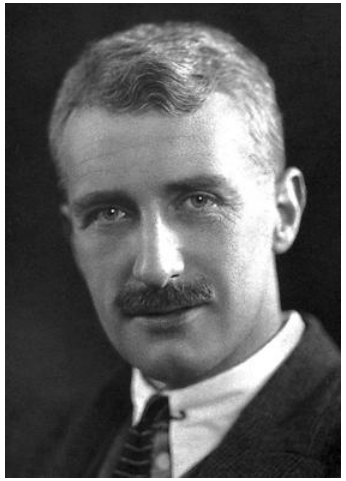
- Muscle fibers are assumed to be straight, parallel, of equal length, coplanar, and attached to the tendon at a certain pennation angle.
- As the muscle shortens, the distance between fibers remains constant and the pennation angle increases.

- Muscle-generated force:
  - Force is a scalar quantity -- tension.
  - Uniform along musculotendon path.
  - Determined by muscle length, muscle velocity, and activation level.

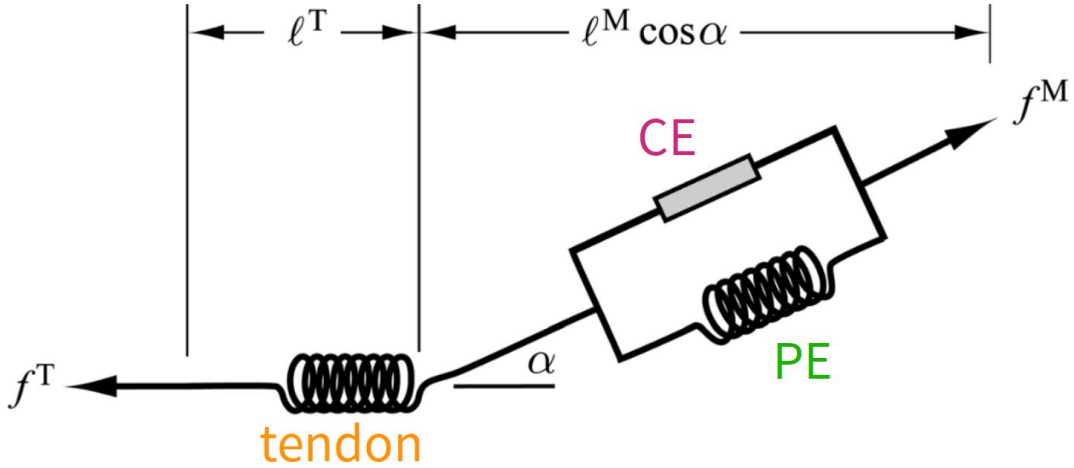


The Hill-Type Muscle Model

- A simplified model for computing muscle forces
- Consists of a contractile element (CE) and a parallel elastic element (PE), in series with an elastic tendon.



Archibald Vivian Hill



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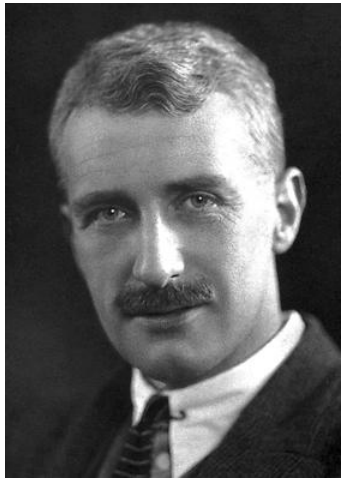

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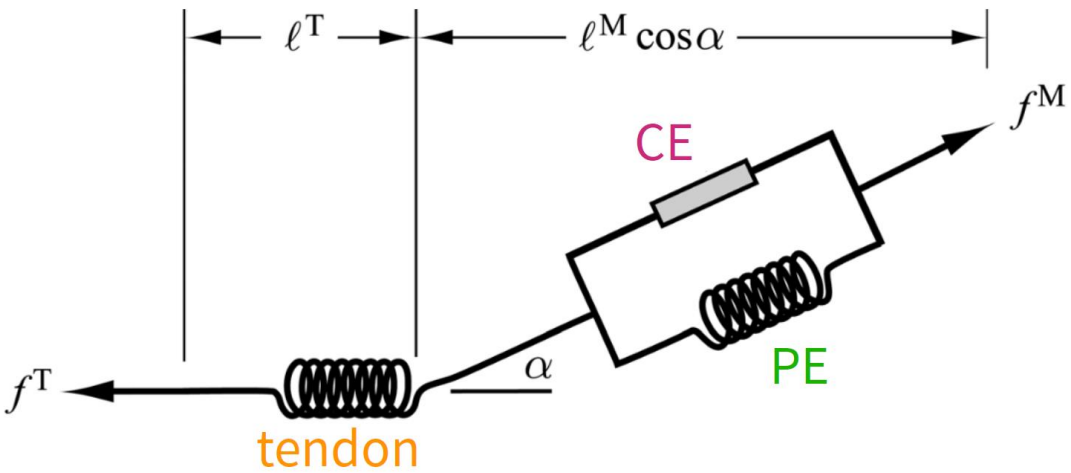
- The muscle force (CE + PE) needs to balance the tendon force:

$$f^T = f^M \cos\alpha$$



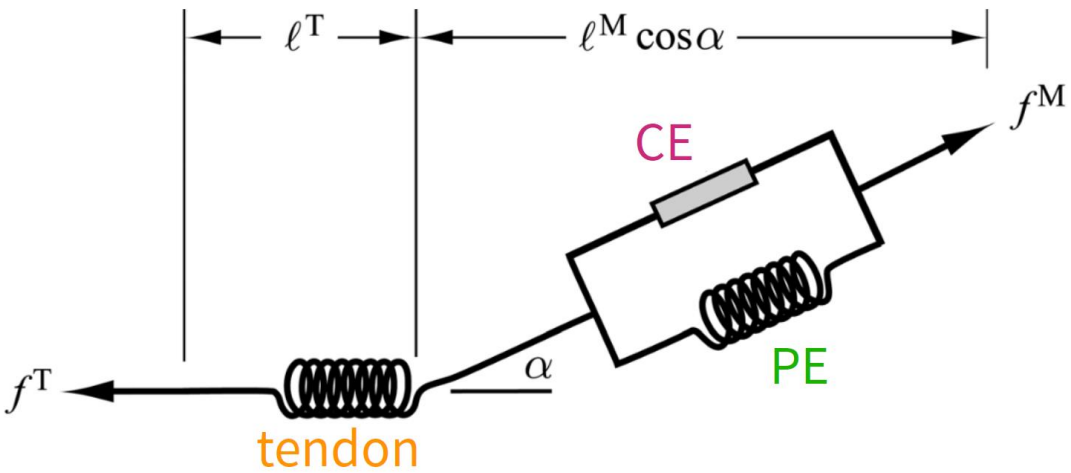
Compute Muscle Force

$$f^T = f^M \cos\alpha$$

$$f^M = f_{CE} + f_{PE}$$

$$f_{CE} = \bar{f}_o \cdot a \cdot f^L(l^M) \cdot f^V(\dot{l}^M)$$

$$f_{PE} = \bar{f}_o f^{PE}(l^M)$$



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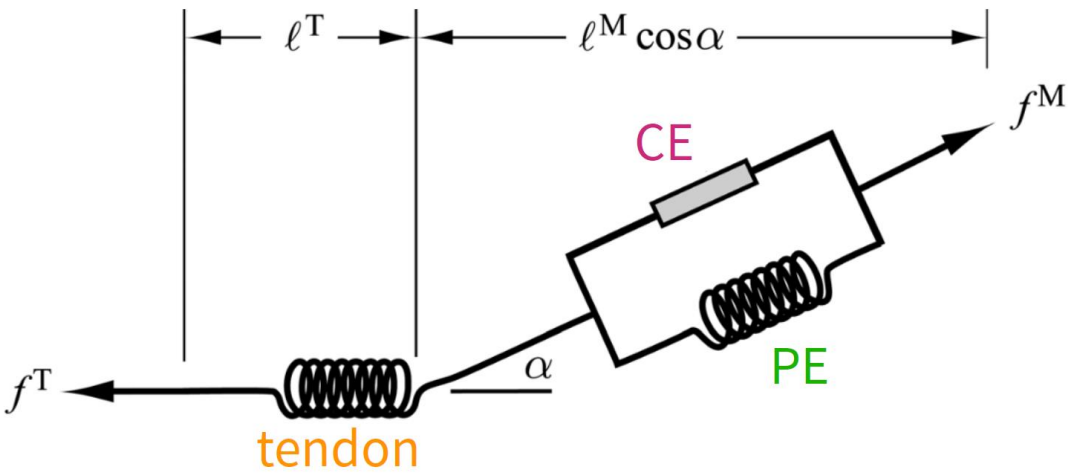
a: actuation level

$$f_{PE} = \bar{f}_o f^{PE}(l^M)$$



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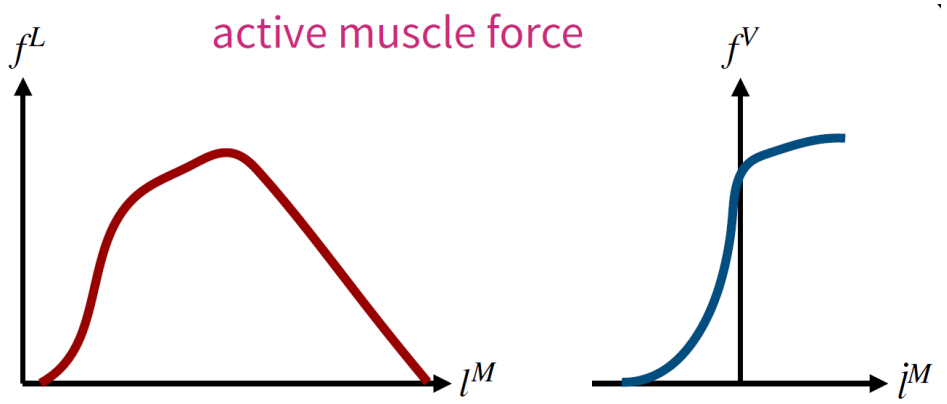

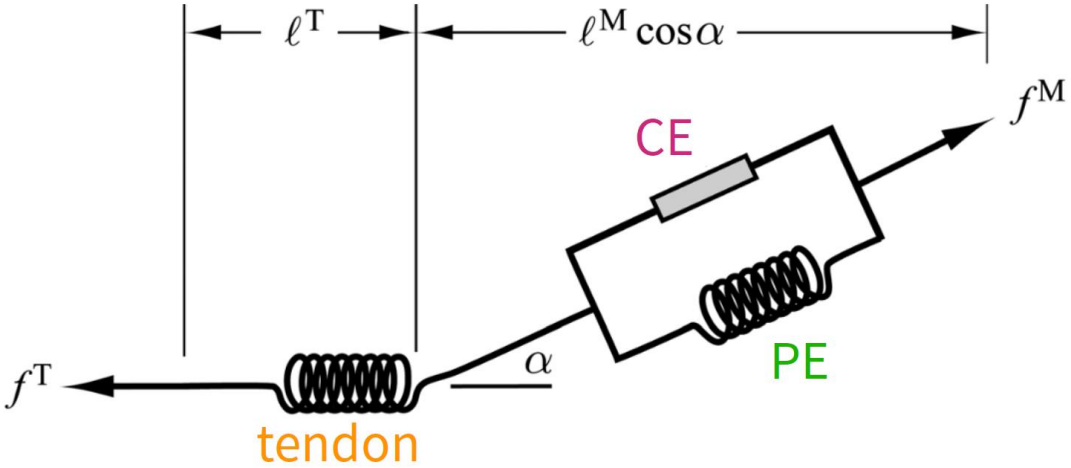

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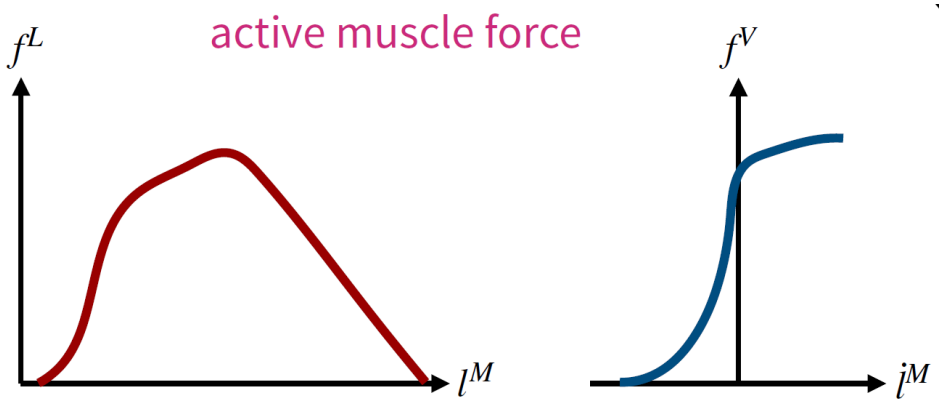

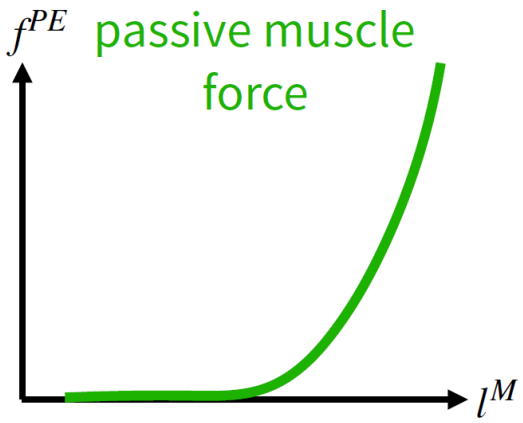

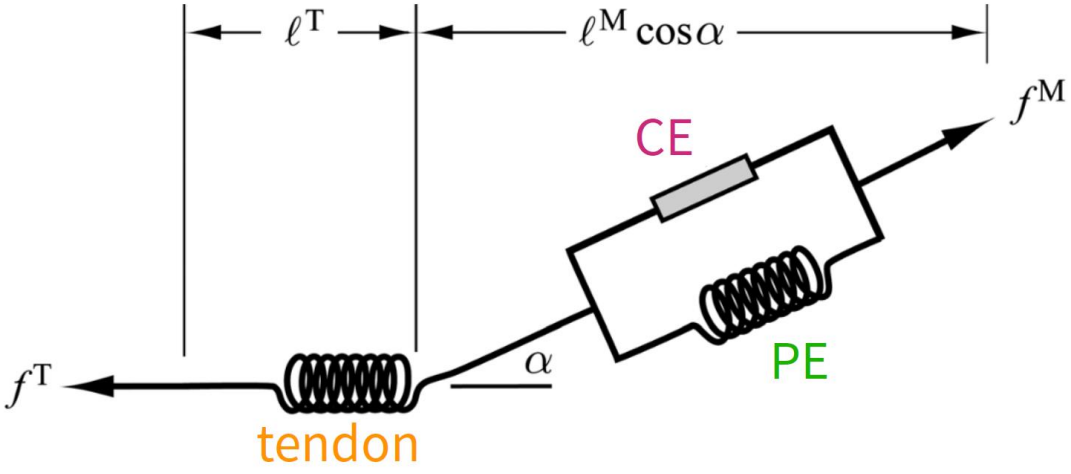

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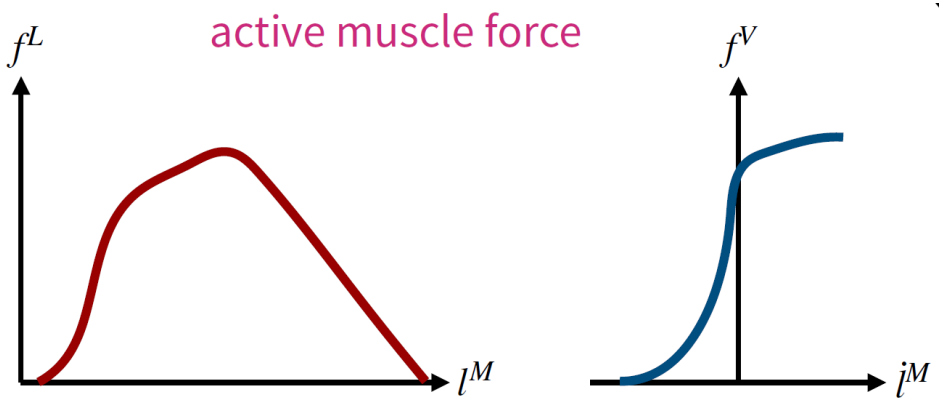
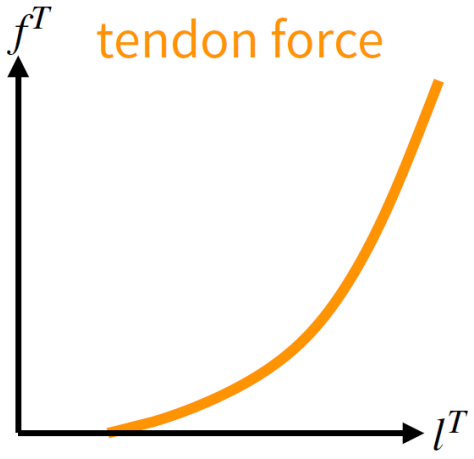
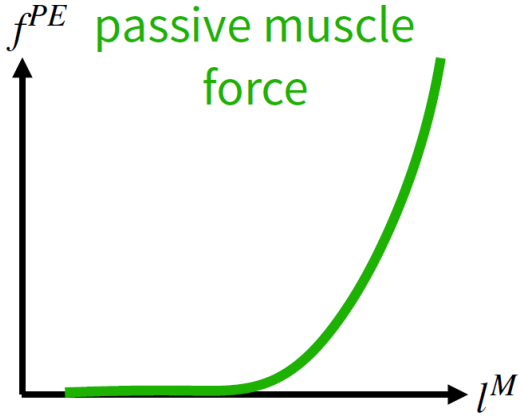
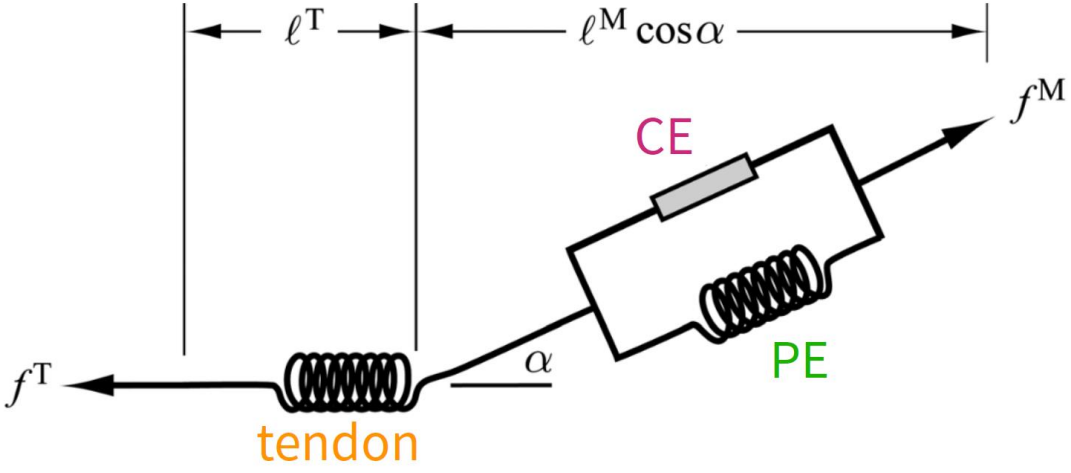








Compute Muscle Force

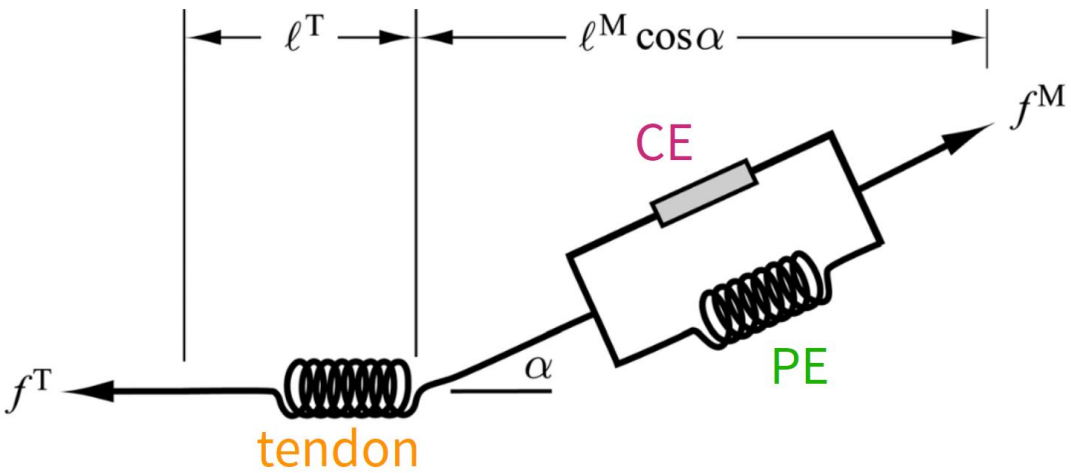
$\bar{f}_o$: muscle normalization constant

a: actuation level

$$f^T = \bar{f}_o \left(a \cdot f^L(l^M) \cdot f^V(\dot{l}^M) + f^{PE}(l^M) \right) \cos\alpha$$

$$f^T = \bar{f}_o \cdot f^T(l^T)$$

$$\bar{f}_o \left(\left(a \cdot f^L(l^M) \cdot f^V(\dot{l}^M) + f^{PE}(l^M) \right) \cos\alpha - f^T(l^T) \right) = 0$$



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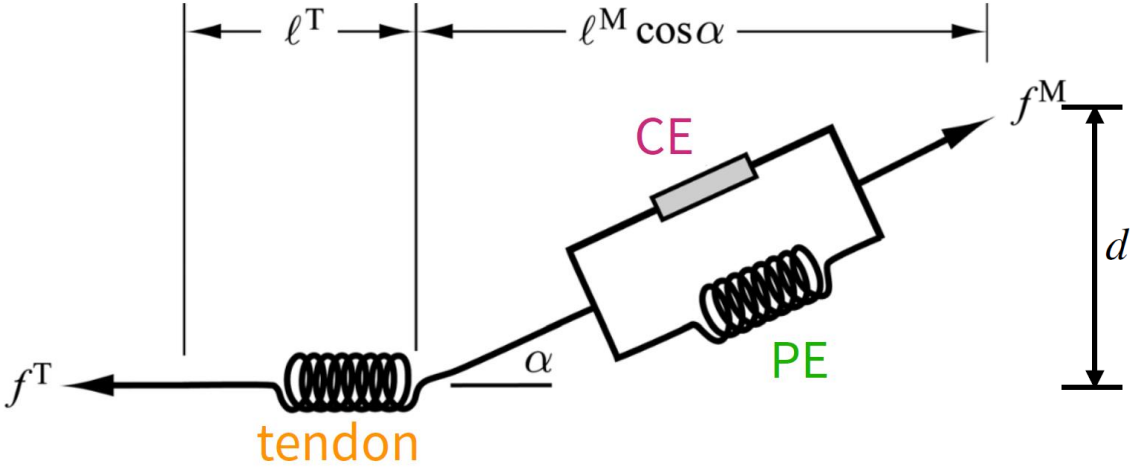
Unknowns: $l^M, \dot{l}^M, l^T$. We need more equations to solve them…



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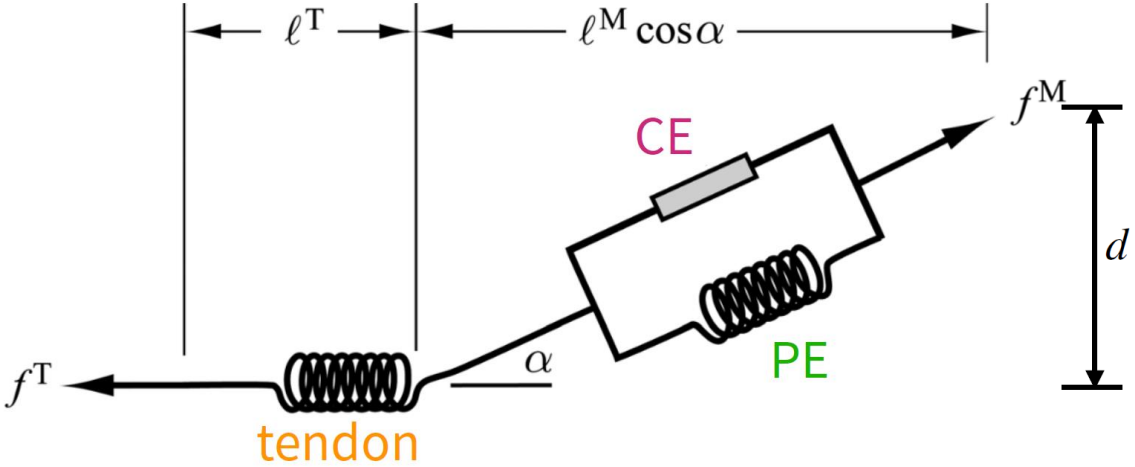
Unknowns: $l^M, \dot{l}^M, l^T$. We need more equations to solve them…

Pennation angle α is also unknown, but can be computed

$$\alpha = \sin^{-1}\frac{d}{l^M}$$



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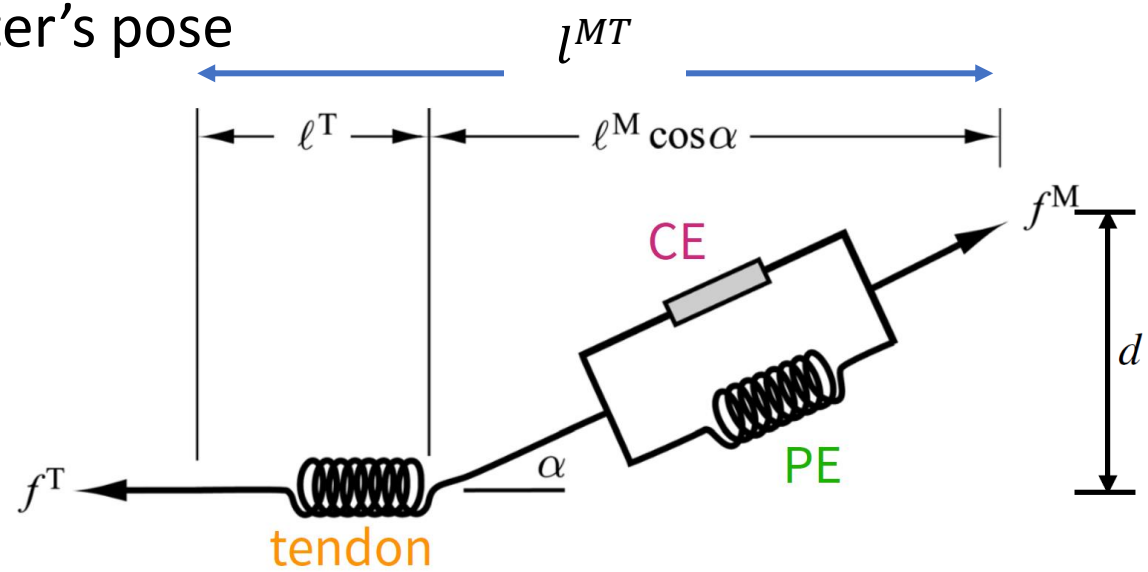
Unknowns: $l^M, \dot{l}^M, l^T$. We need more equations to solve them…

Let $l^{MT} = l^T + l^M \cdot \cos\alpha$

$$\alpha = \sin^{-1}\frac{d}{l^M}$$

l^{MT} can be computed from the character's pose

We still need one extra equation.



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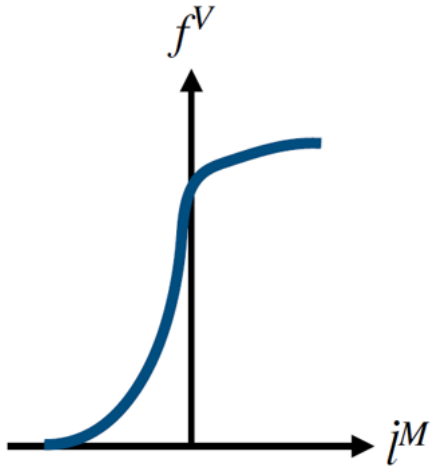
$$\bar{f}_o\left(\left(a \cdot f^L(l^M) \cdot f^V(\dot{l}^M) + f^{PE}(l^M)\right)\cos\alpha - f^T(l^T)\right) = 0$$

$$\dot{l}^M = f^V_{inv}\left(\frac{f^T(l^T) - f^{PE}(l^M)\cdot\cos\alpha}{a \cdot f^L(l^M)\cdot\cos\alpha}\right)$$

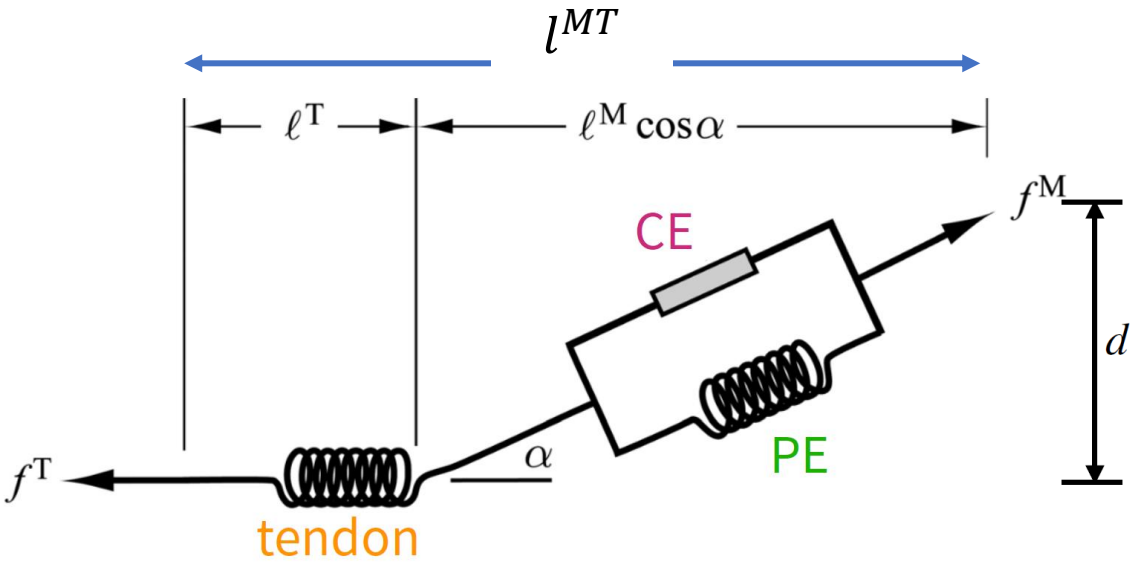


$$\alpha = \sin^{-1}\frac{d}{l^M}$$

We can track l^M by integrate this ODE:

$$l^M_{t+1} = l^M_t + h \cdot \dot{l}^M$$

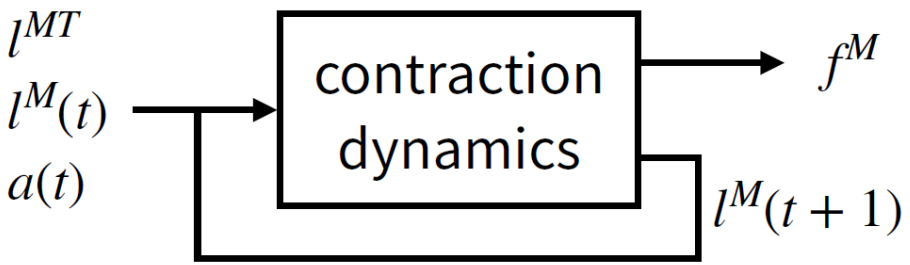
$$= l^M_t + h \cdot f^V_{inv}(***)$$

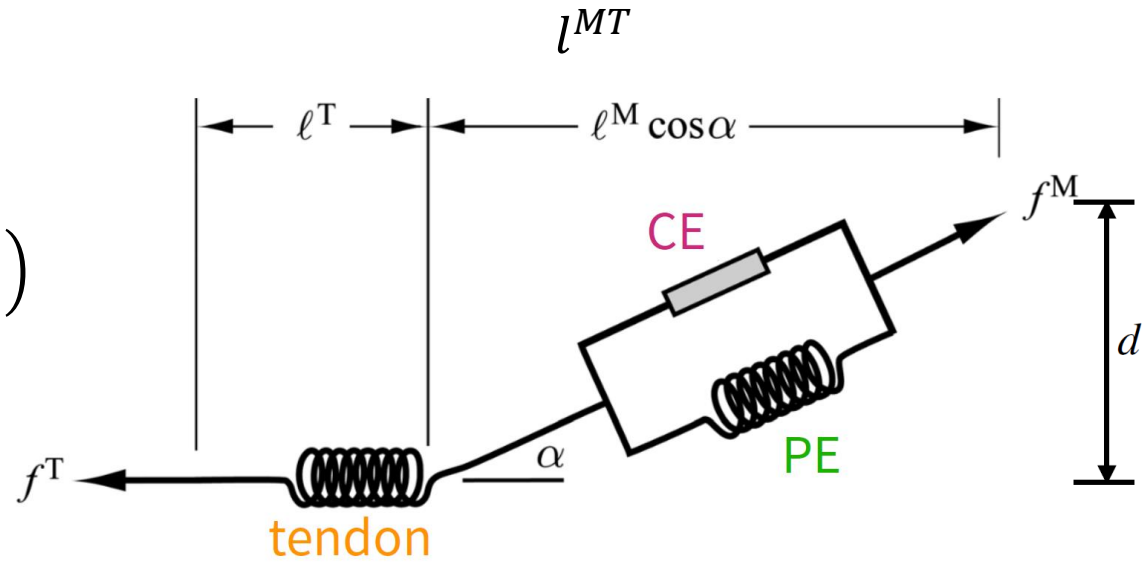


Muscle Contraction Dynamics

Given l_0^M, at time t, given a_t:

0. Compute l^{MT} based on character's pose
1. $\alpha = \sin^{-1}\frac{d}{l_t^M}$
2. $l^T = l^{MT} - l_t^M \cdot \cos\alpha$
3. $\dot{l}^M = f_{inv}^V\left(\frac{f^T(l^T) - f^{PE}(l_t^M)\cdot\cos\alpha}{a_t \cdot f^L(l^M)\cdot\cos\alpha}\right)$
4. $f^M = \bar{f}_o\left(a_t \cdot f^L(l^M)\cdot f^V(\dot{l}^M) + f^{PE}(l^M)\right)$
5. $l_{t+1}^M = l_t^M + h\cdot\dot{l}^M$





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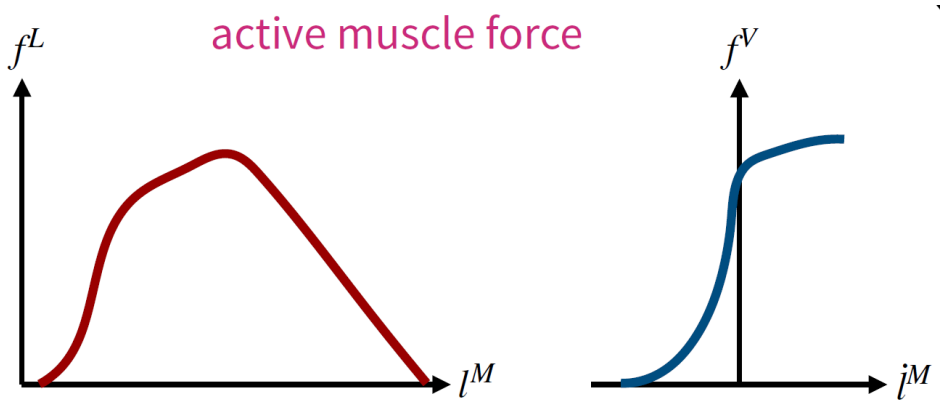
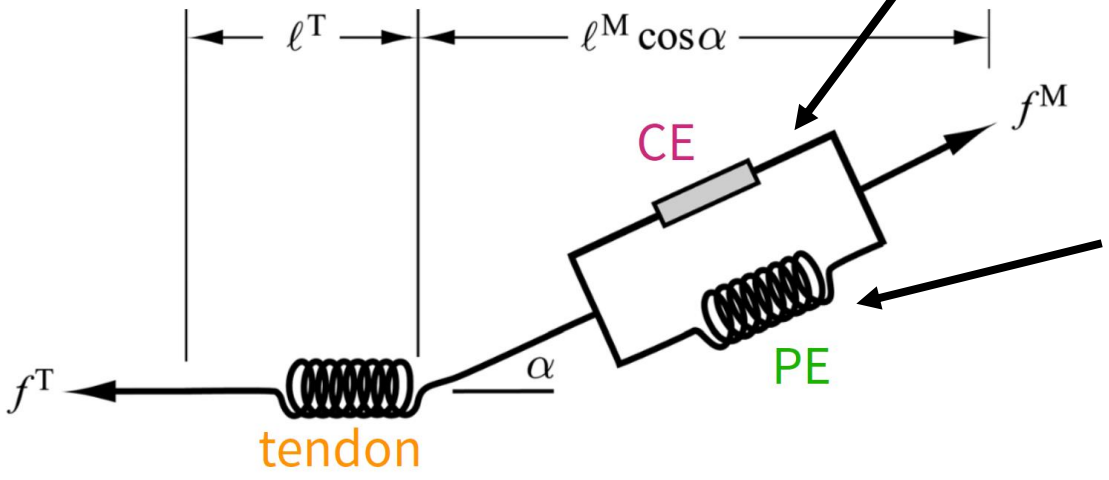
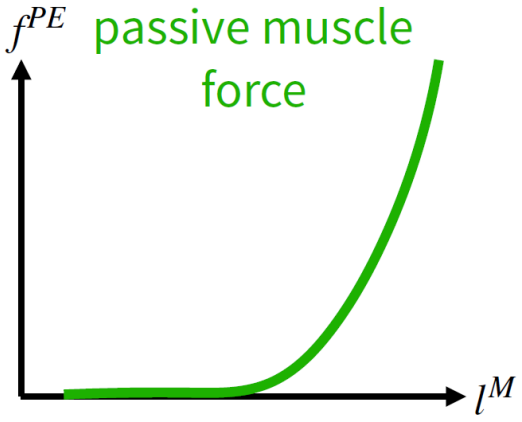


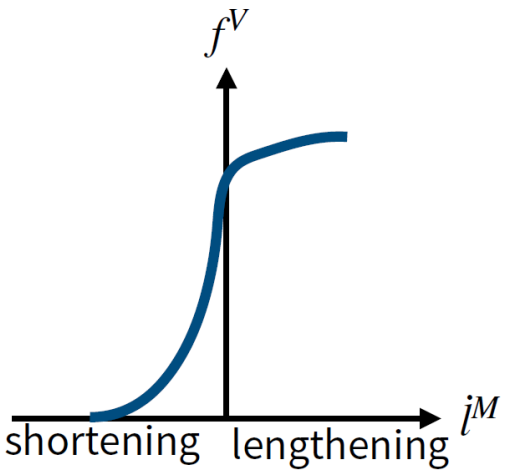


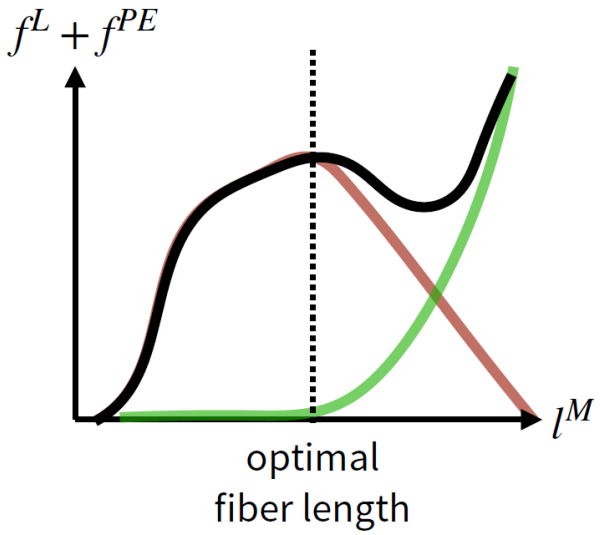


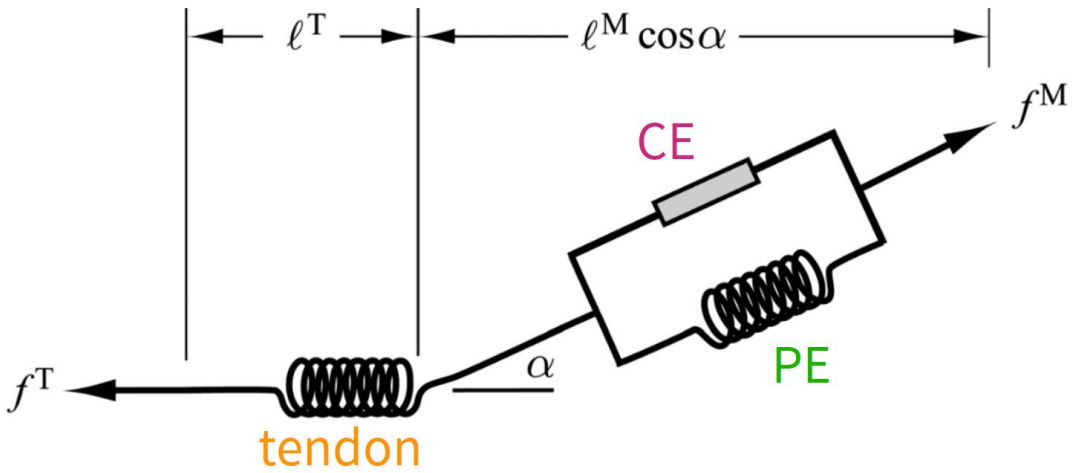
Compute Muscle Force

$$f^M = \bar{f}_o \left(a_t \cdot f^L(l^M) \cdot f^V(\dot{l}^M) + f^{PE}(l^M) \right)$$

- Every muscle has an optimal fiber length. Either hyper flexion or hyper extension reduces the actuation ability.
- f^V reduces muscle activation when muscle is shortening, providing intrinsic stability to avoid high shortening speed.





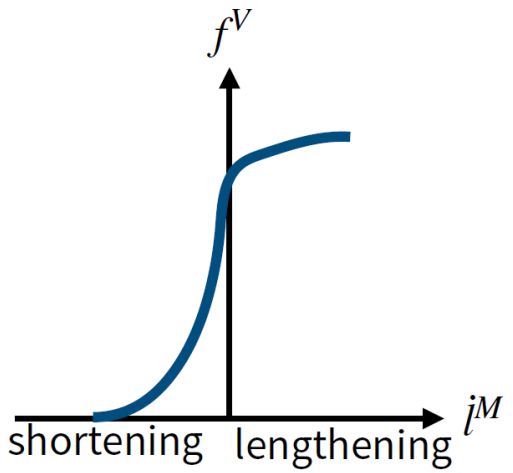


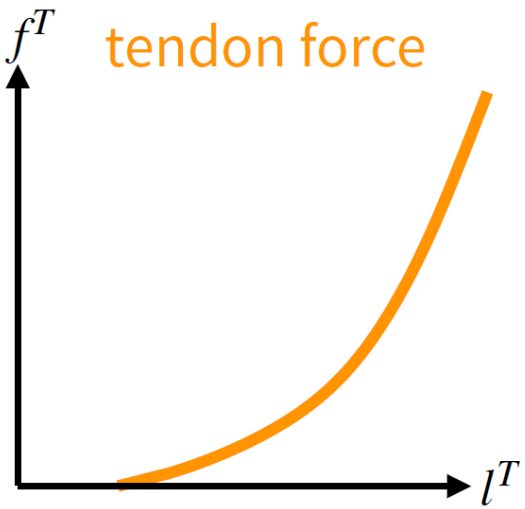
Compute Muscle Force

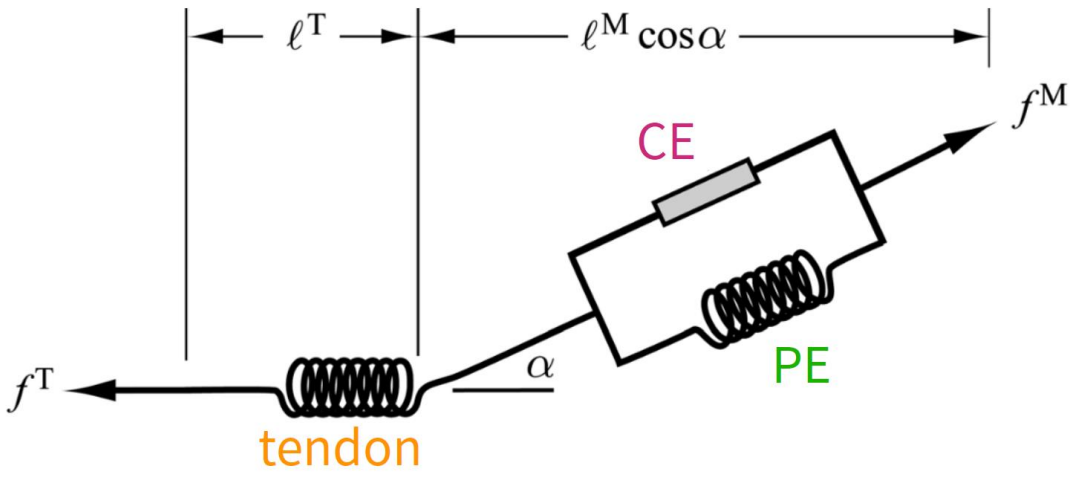
$$f^M = \bar{f}_o \left(a_t \cdot f^L(l^M) \cdot f^V(\dot{l}^M) + f^{PE}(l^M) \right)$$

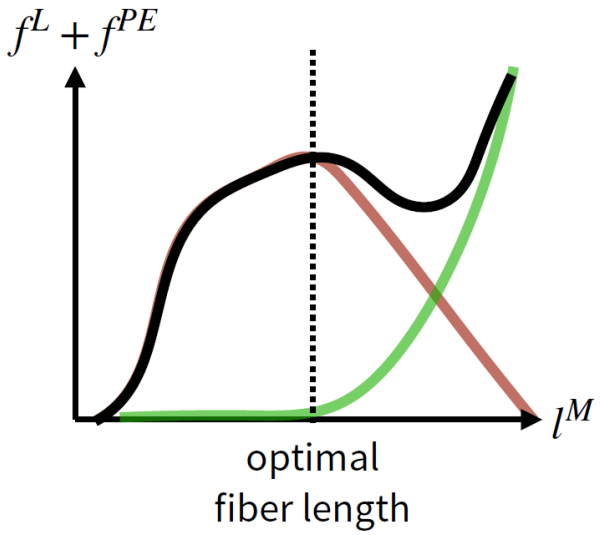
$$f^T = \bar{f}_o \cdot f^T(l^T)$$

- Tendon force: a nonlinear "spring" that stores elastic energy when stretched beyond slack length.







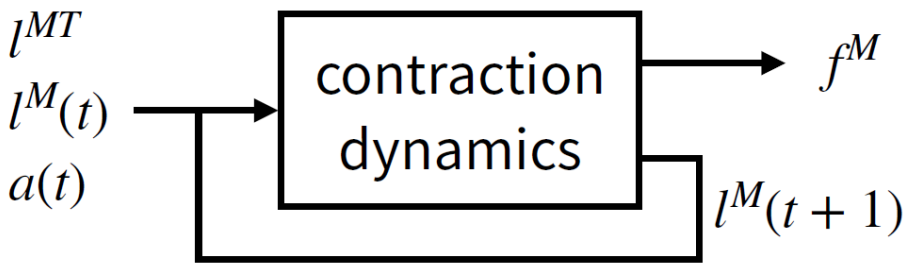


Activation Dynamics

- In contraction dynamics model, we have assumed that the activation $a \in [0,1]$ can change for each timestep
  - This can cause problem if a change too

$$\bar{f}_o\left(\left(a \cdot f^L(l^M) \cdot f^V(\dot{l}^M) + f^{PE}(l^M)\right)\cos\alpha - f^T(l^T)\right) = 0$$

$f^V(\dot{l}^M)$ might need to be very large to balance the tendon force.



Activation Dynamics

- In real human musculoskeletal system, activation cannot change arbitrarily fast
  - $\tau_A = 10\ ms$ to activate
  - $\tau_D = 40\ ms$ to deactivate

$$\dot{a} = \frac{u - a}{\tau}$$

a is the current activation value
$u \in [0,1]$ is the neural excitation command

$$\tau = \begin{cases} \tau_A(0.5 + 1.5a) & \text{, if } u > a \\ \frac{\tau_D}{0.5 + 1.5a} & \text{, otherwise} \end{cases}$$

If a is large, $\dot{a}$ is small. That is, activation is slower when the activation level is already high.

If a is large, $\dot{a}$ is large. That is, deactivation is faster when the activation level is high.

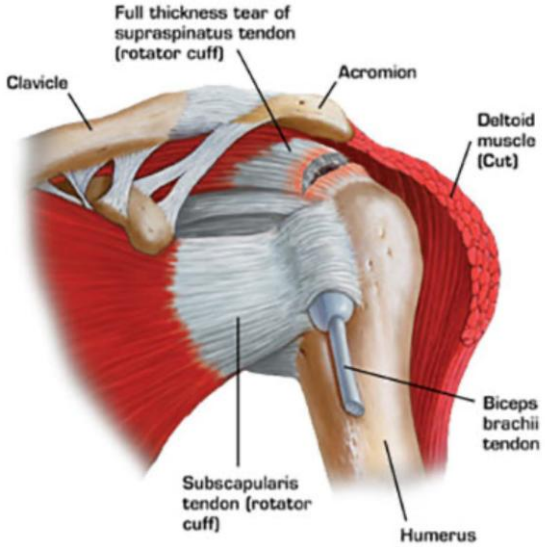
Rigid Tendon Simplification

In previous computations, we have:

$$\bar{f}_o\left(\left(a \cdot f^L(l^M) \cdot f^V(\dot{l}^M) + f^{PE}(l^M)\right)\cos\alpha - f^T(l^T)\right) = 0$$

$$l^T + l^M \cdot \cos\alpha = l^{MT}$$

Unknowns: $l^M, \dot{l}^M, l^T$.



However, tendons are in general stiff compared with muscle fibers.

We can simplify contraction dynamics by treating tendon as an inextensible wire with a fixed length $\bar{l}^T$

Now we do not need to keep a record of l^M, because $l^M, \dot{l}^M$ can be computed by $l^{MT}, \dot{l}^{MT}$, which can be computed from the character's state

Rigid Tendon Simplification

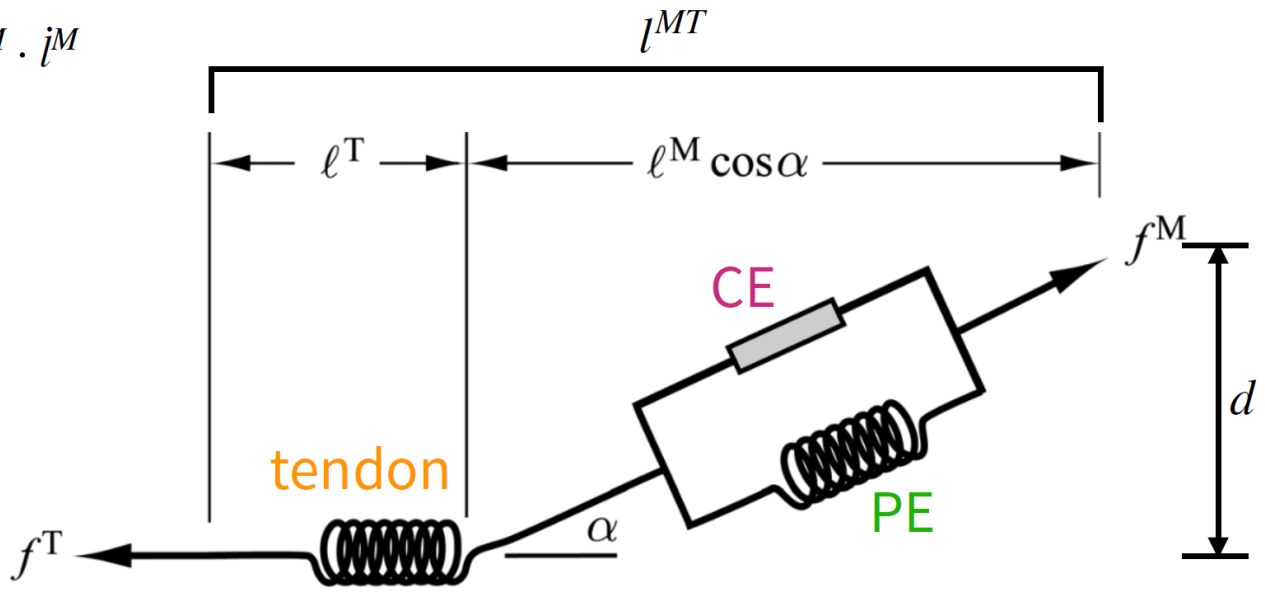
Given l^{MT} and $\dot{l}^{MT}$ and the rigid tendon length $\bar{l}^T$, how to compute the muscle length l^M and muscle velocity $\dot{l}^M$?

$$l^M = \sqrt{d^2 + (l^{MT} - \bar{l}^T)^2}$$

$$\dot{l}^{MT} = \dot{l}^T + \frac{d}{dt}(l^{M^2} - d^2)^{1/2} = \frac{1}{2}(l^{M^2} - d^2)^{-1/2} \cdot 2l^M \cdot \dot{l}^M$$

$$= \frac{\dot{l}^M}{\cos\alpha}$$

$$\dot{l}^M = \dot{l}^{MT} \cdot \cos\alpha = \dot{l}^{MT} \cdot \sqrt{1 - (\frac{d}{l^M})^2}$$

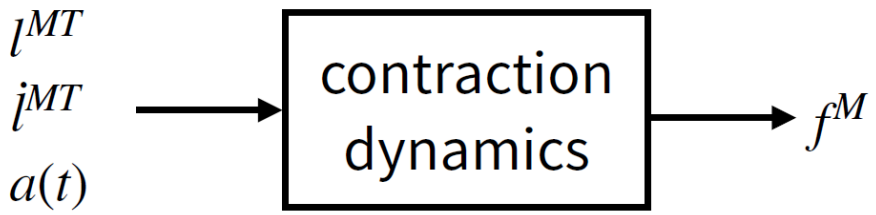


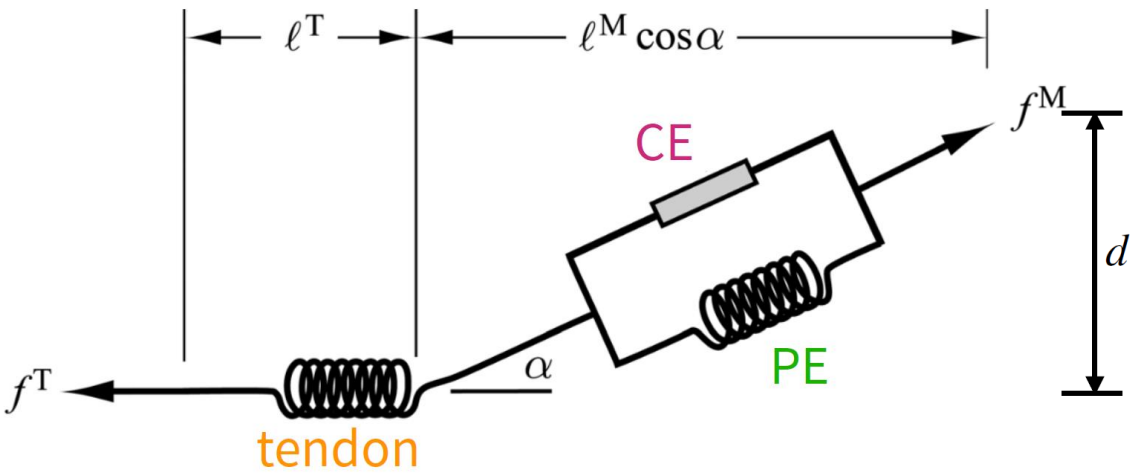
Rigid Tendon Simplification

1. $l^M = \sqrt{d^2 + \left(l^{MT} - \bar{l}^T\right)^2}$

2. $\dot{l}^M = \dot{l}^{MT} \cdot \sqrt{1 - \left(\frac{d}{l^M}\right)^2}$

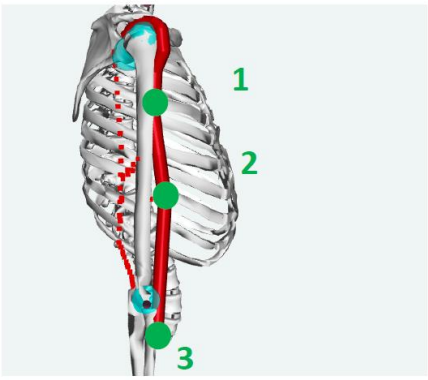
3. $f^M = \bar{f}_o \left(a_t \cdot f^L(l^M) \cdot f^V(\dot{l}^M) + f^{PE}(l^M)\right)$

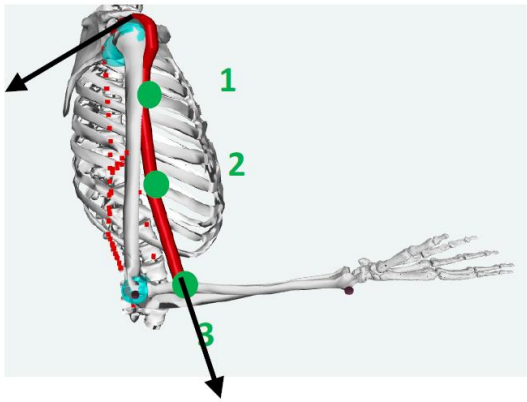


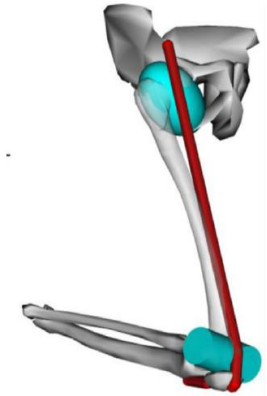


Muscle Path

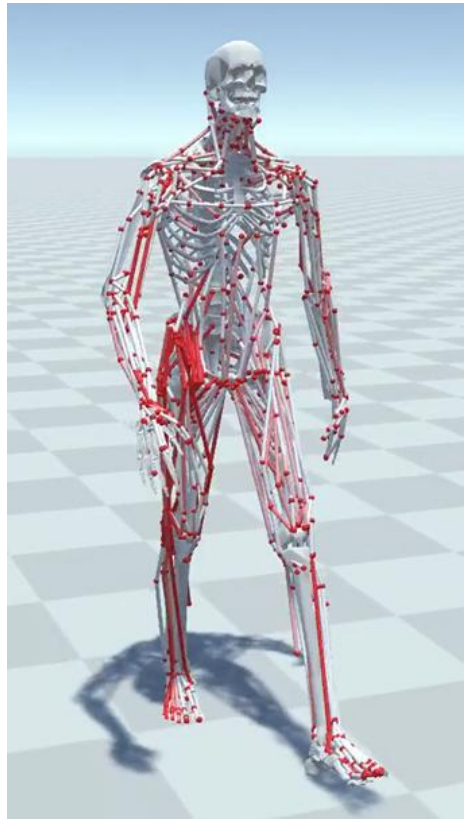
- Muscles can span several fixed points relative to the bones.
- Muscles can also wrap and slide around joints.
- Muscle paths can be computed from skeletal state
- The directions of muscle force are determined by the tangents at the two end points of muscle path.
  - How to apply this force to the bone??

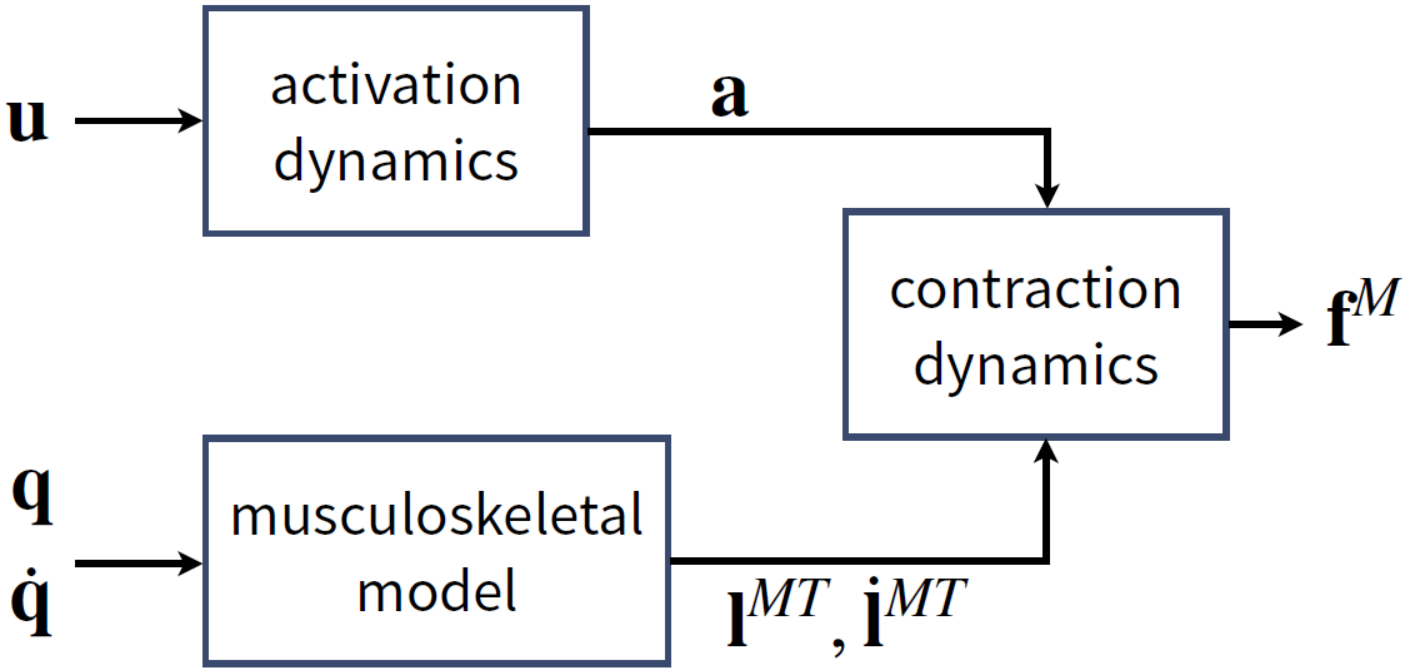




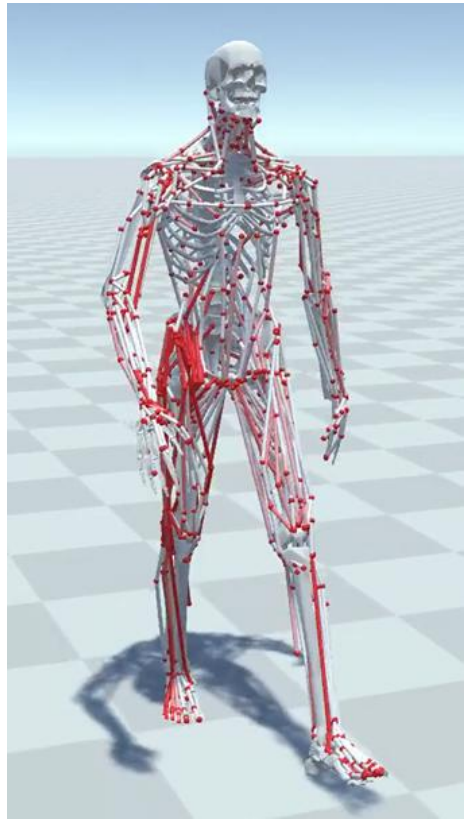


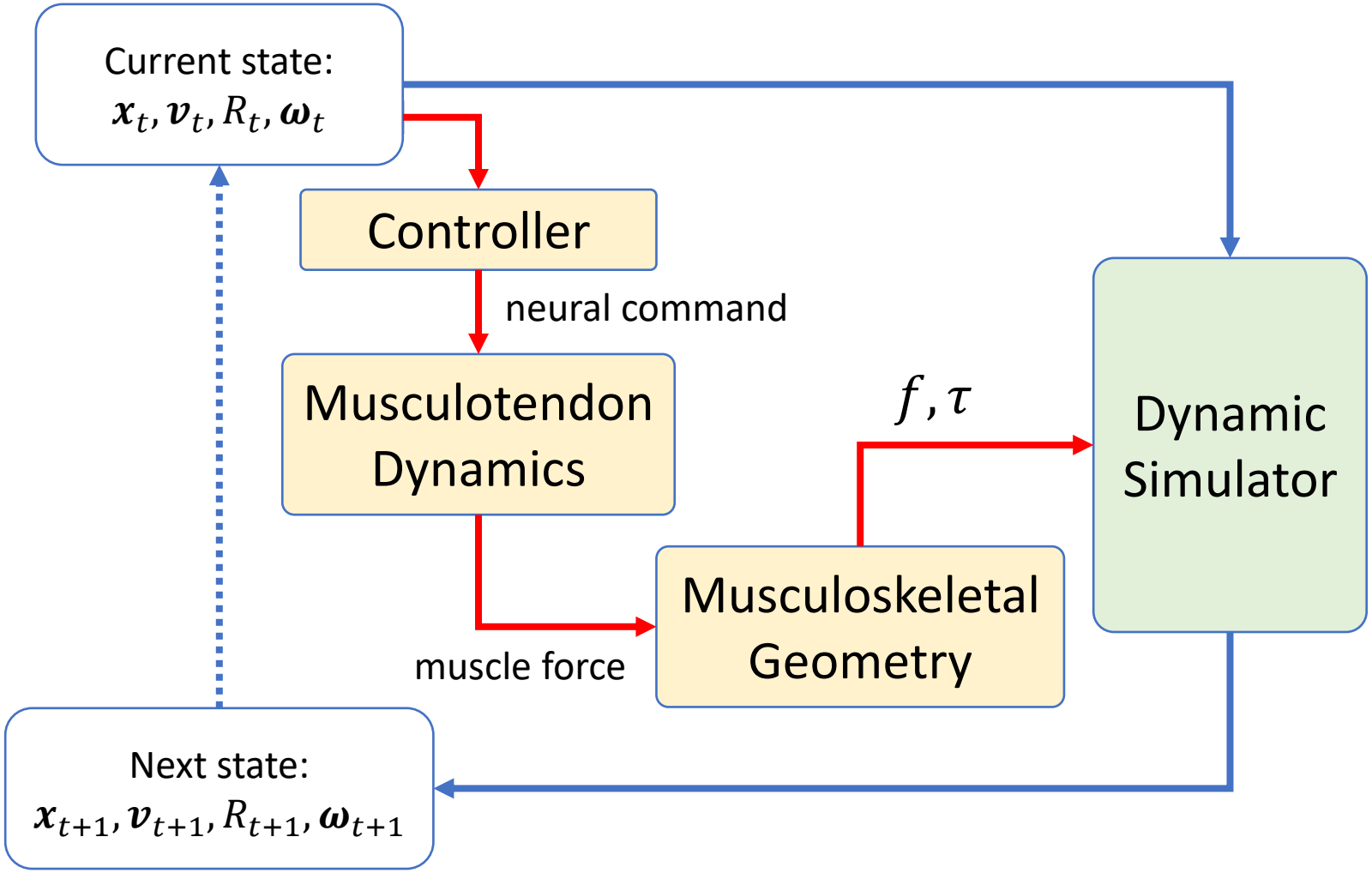
Musculotendon Dynamics





Simulating + Controlling a Character





OpenSim: A simulation framework for musculoskeletal systems

The proposed spinal control modules are sufficient to generate steady and transitional locomotion behaviors

slope ascend and descend

https://simtk.org/projects/opensim

Muscle VAE

Motion Tracking

Questions?